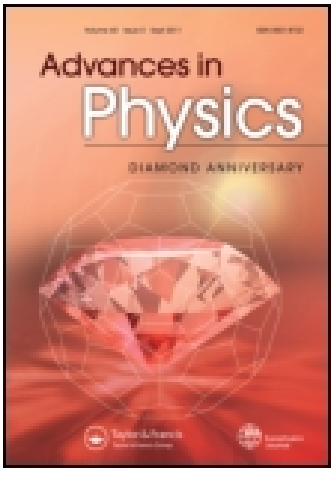
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Studies of Remanent Magnetism and Viscous Magnetism in the Basalts of Iceland

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§ 1. INTRODUCTION

THE paper gives a brief description of the instrument which the author constructed for measuring the direction and intensity of the magnetic moments. It gives further the most important results of the studies of demagnetization in an alternating field, and of the investigation of viscous magnetization. The paper describes also an investigation of the secular variation in Iceland during the last 5000 years. Finally it describes the results of the investigation of an area between normally and reversely magnetized zones, that is, it describes how the direction of magnetization changes continuously from reversed to normal.

§ 2. THE INSTRUMENT

Basalts are somewhat inhomogeneously magnetized. Koenigsberger (1932) showed that when he cut a 4 by 4 by 4 cm sample into eight 2 by 2 by 2 cm samples, the spread in the direction of magnetization of the smaller samples was very great (sometimes it amounted to as much as 15°). Therefore it is of interest to make measurements on large samples ; at the same time the method must be such that we measure the vectorial sum of the individual magnetic moments in the sample. That means the output must be proportional to

$$\iiint \boldsymbol{\mu}(x, y, z)\, dx\, dy\, dz$$

integrated over the entire sample ; where $\boldsymbol{\mu}$ is the magnetic moment per unit volume.

The instrument which we constructed fulfilled this requirement and with it we can perform measurements on any sample smaller than about 1200 cm^3.

The instrument is an induction instrument. The sample rotates with a frequency of 5 c.p.s. The induced current is commutated and then measured with a galvanometer. The sensitivity is 10^{-7} e.m.u./g., it is limited by the commutator noise. The specimen is fixed in a Cardan-suspension. This device makes it possible to install the specimen with its direction of magnetization accurately parallel to the axis of rotation. In this way the direction of magnetization is easily determined with a 0·1° accuracy (*zero-method*). The constant earth-field is reduced by 1/100 or to 0·005 Oersted by means of mu-metal screens. The mu-metal screen, a condenser in the circuit and the stand, that is a good conductor, reduce the fluctuations by more than 1/600. The fluctuations are also reduced by the commutator.

The instrument is portable.

§ 3. Demagnetization

Preliminary studies indicated that partial demagnetization is of great interest, as the intensity of the normally magnetized basalts decreased for every increase of the demagnetizing field ; while the intensity of reversely magnetized samples of basalt *increased* as they were partially demagnetized in alternating field smaller than 30–50 Oe. For greater demagnetization fields, the intensity decreased as for normally magnetized basalts. Further preliminary studies showed that the viscous magnetization was removed by a partial demagnetization of the sample (Brynjólfsson 1956).

We represent the demagnetization effect by means of negative differential demagnetization curves. That means the differential quotient

$$(-dI/dH_{\max})$$

is plotted as a function of $H_{\max}$. Here I is the intensity of the magnetization of the sample and $H_{\max}$ the maximal demagnetization field

From fig. 1 we see that the differential demagnetization curves are very different for different samples.

Curves one and two belong to postglacial basalts (which are, of course, normally magnetized), curve three to a normally magnetized tertiary basalt, and curves four and five to a reversely magnetized tertiary basalt. It is especially interesting that with reversely magnetized basalt, the intensity always increases at the beginning. (Experimental result I.) When the sample is strongly magnetized, the effect is often compensated by the ordinary decrease by demagnetization.

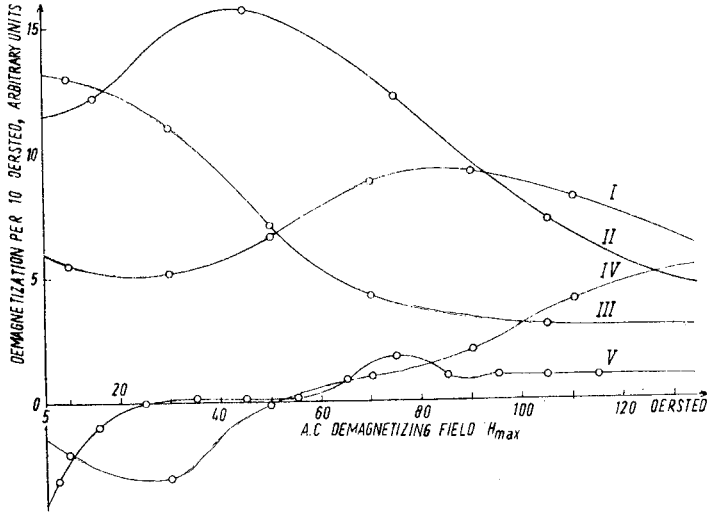
The average of many (14) differential demagnetization curves for postglacial basalts is nearly constant. It decreases just a little at the end. (Experimental result II.)

The amount of the decrease of the intensity depends on the angle between the direction of the magnetization of the sample, and the direction of the alternating field, as follows. (Experimental result III.)

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When the postglacial basalt sample is demagnetized in all directions with $H_{max}=140$ Oe, the intensity decreases from 100 to 57, i.e. if $I^0=100$ we have $I_*^{140}=57$, a change of $43\%\pm1{\cdot}2\%$. The decrease of the intensity fluctuates between 63% and 9%, and the circle of confidence for each sample is 12°. When the sample is demagnetized in all directions perpendicular to its direction of magnetization, the intensity decreases from 100 to 66·6, i.e. if $I^0=100$ $I_\perp^{140}=66{\cdot}6$. This can be done by rotating the sample about the axis defined by its direction of magnetization, with a demagnetizing field perpendicular to this axis. If we next rotate a new sample about an axis perpendicular to its direction of magnetization, keeping the direction of the demagnetizing field constant, and therefore demagnetizing in all directions in a plane perpendicular to the preceding plane containing the magnetization vector of the sample, the intensity decreases from 100 to 59·6, i.e. if $I^0=100$ we have $I_\pm^{140}=59{\cdot}6$. If both of these operations are performed on the same sample, thereby demagnetizing in all directions, the intensity decreases from 100 to 57, as previously stated.

Fig. 1



The experimental results II and III of differential demagnetization together with the experimental results of Schmucker (1956) which show that the thermo-remanent magnetization is proportional to the external field (when the external field is lower than about 10 Oe)—coupled further with the hypothesis that the sample contains many grains or domains which can be magnetized in *two anti-parallel directions only*, and that the probabilities for this direction of easy magnetization to be in one or another direction are equal—make it possible to calculate how the amount of demagnetization depends on the direction of H_{max}.

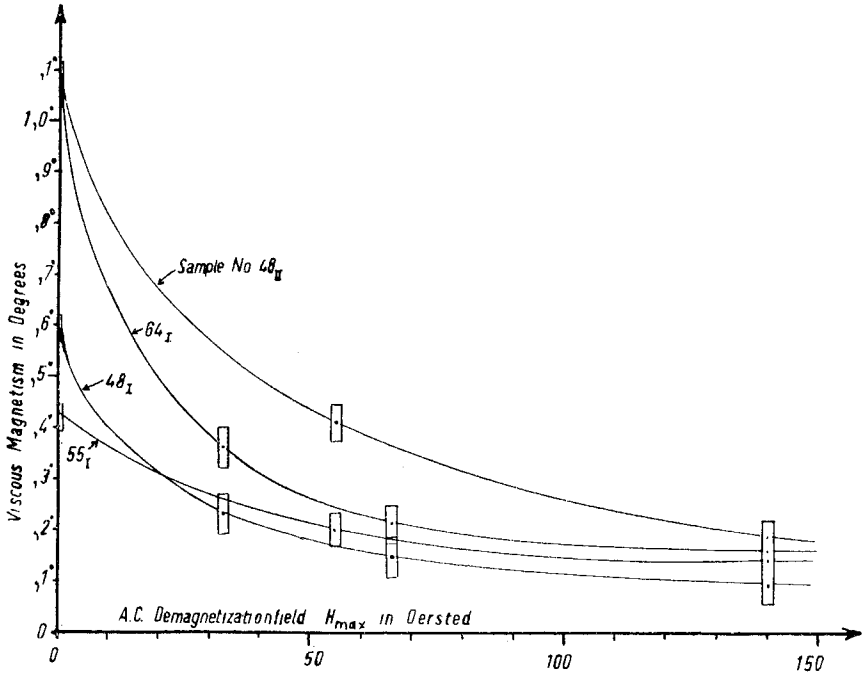
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The values calculated in this manner are in accordance with the experimental ones. If $I^0=100$, then the values of I_*^{140}, $I_\pm^{140}$ and $I_\perp^{140}$ are as follows :

	Experimental values	Calculated values
I_*^{140}	57·0	(57)
$I_\pm^{140}$	59·6	60·6
$I_\perp^{140}$	66·6	67·0

I.e. we can explain the experimental results when we assume that there only exist two directions of easy magnetization in each grain; on the other hand the experimental results are in contradiction to the hypotheses that there exist many more directions of easy magetization in each grain. Further as we have the good agreement between the experimental and the calculated values, we must assume that the anisotropy in each grain is very strong.

Fig. 2



§ 4. Viscous Magnetism

As we can measure the change of direction of the magnetic moment very accurately, we can easily investigate the viscous magnetization.

At the beginning, we demagnetize the sample in all directions with 140 Oe and determine the direction of the magnetic moment with 5'

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accuracy. Then we fix the sample with its direction of magnetization at a right angle to the direction of the earth field. The viscous magnetization is then proportional to the change in direction. This change after some months amounts to 1° to 10° which is large compared with 5′.

It is now of great interest that the *viscous magnetization is mostly removed when the sample is partly demagnetized in alternating fields* (H_{max}=*ca.* 100 Oe). (Experimental result IV.)

Figure 2 illustrates how strongly the viscous magnetism is reduced by partial demagnetization.

The ordinate shows the amount of viscous magnetization and the abscissa the maximal demagnetization field H_{max}.

The laboratory investigation of viscous magnetism thus indicates that in case of reversely magnetized basalts the increase of intensity on demagnetization with H_{max}=*ca.* 40 Oe is due to the removal of the viscous magnetism. (Experimental result V.) This then indicates that in the case of tertiary basalts the viscous magnetism (caused by the present normal field) amounts to 5×10^{-4} to 5×10^{-5} e.m.u./g. In the case of weakly magnetized samples this is a very serious effect. It is then of great interest to us that this viscous magnetism can be removed (mostly) by partial demagnetization with H_{max}=100 Oe. (The sample can be cleaned.)

In the theory of viscous magnetization worked out mainly by Street and Woolley (1949), Street, Woolley and Phillips (1954) and Street, Woolley and Smith (1952) and, in a different way, by Néel (1949, 1950, 1951, 1952) the viscous magnetism is thought of as caused by the thermal fluctuations. Thus the thermal fluctuations can bring a domain from one state to another ; the thermal fluctuations deform the domains and, as the magnetic state depends on the form of and strains in the grains, there exists a coupling between the thermal fluctuations and the magnetic state.

The influence of thermal fluctuations is described by the Maxwell–Boltzmann term $\exp\{-(E/kT)\}$.

If we assume with Street and Woolley that the energy E is nearly equal to $I_s \,.\, v \,.\, H_c$, we can find the volume v of the domains from the form of demagnetization curves of the viscous magnetization. This gives a diameter for the domains equal to or greater than 2×10^{-6} cm. (Experimental result VI.)

If we use Néel's expression for E, we find almost the same volume for the domains. That is, the investigation of viscous magnetism indicates that the *diameter of the smallest domains is about* 2×10^{-6} *cm.*

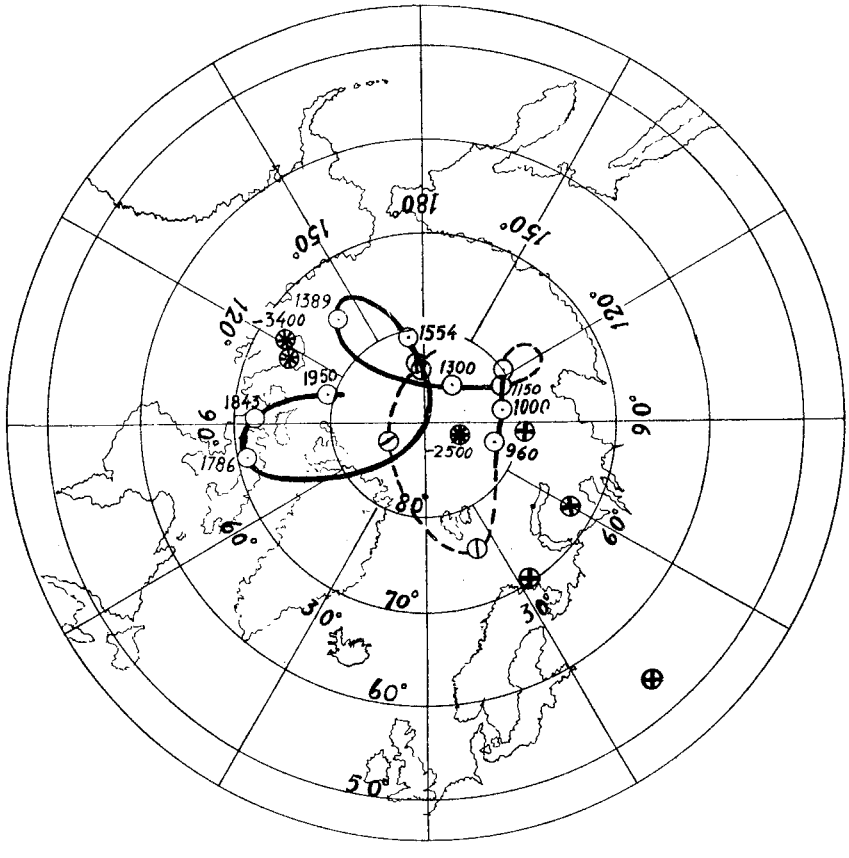
§ 5. Postglacial Basalts

Twenty-five postglacial lavas were investigated. The intensity of magnetization I_r in lava-flows, which cooled down after A.D. 500, fluctuated between $1{\cdot}2\times10^{-2}$ and $1{\cdot}1\times10^{-3}$ e.m.u./g. with a mean of $5{\cdot}1\times10^{-3}$ e.m.u./g. (Experimental result VII.) The intensity in the older flows was about 20% smaller.

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Figure 3 shows the secular variation in Iceland. (Experimental result VIII.) The points represent the dipoles that correspond to the direction of magnetization (in Iceland). The positive numbers at the points indicate the age A.D. Those with minus indicate the age B.C. A line across the circle of confidence shows that the age of the corresponding flow is between A.D. 300 and about A.D. 900. A cross in the circle of confidence indicates that the age of the corresponding flow is between 1000 B.C. to about A.D. 200. These points between

Fig. 3

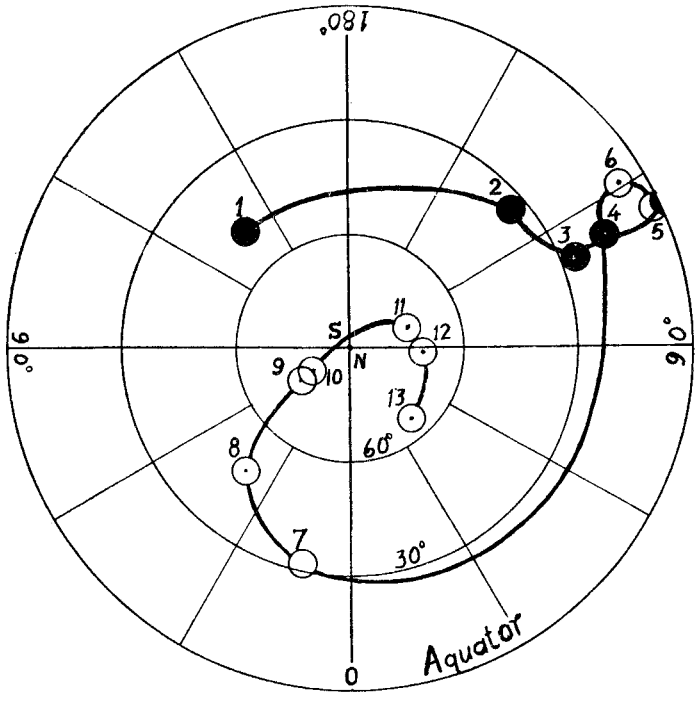


1000 B.C. and about A.D. 200 all show very easterly declination. Although the field in Iceland alone does not define the dipole very accurately, the results indicate that the south pole of the whole field must have been somewhere in the neighbourhood of northern Norway or Novaja Zemlja, U.S.S.R. If this is right, *the ancient Greeks should have seen the northern light much more frequently than they are seen in that region today.* This is in accordance with Aristoteles' exhaustive description in his book "Meteorologia", Chapter IV and V.

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We see that the secular variations are very great. We also see that the mean direction of the field corresponds to a magnetic south pole in the neighbourhood of the geographic north pole and not with the present magnetic south pole. Thus, these measurements indicate *that even during the last 5000 years, the mean dipole field is that corresponding to the magnetic pole coincident with the geographic pole and not corresponding to the dipole of today.* (Experimental result IX.) This is an interesting result as the magnetic direction in Iceland is a sensitive indicator for which of the *two* directions is the more significant.

Fig. 4



§ 6. Tertiary Basalts

It was found that *the direction of magnetization in each lava was much more homogeneous after a partial demagnetization of the sample in alternating field with maximal amplitude* $H_{max} = 140$ Oe, *than before the demagnetization.* (Experimental result X.) This was most clearly seen with weakly magnetized basalts. On partial demagnetization, the weakly demagnetized samples changed their direction much more than the strongly magnetized samples in the same lava. During the demagnetization all the samples from a single lava changed their direction in the same way (not randomly). *At low demagnetization fields* (*ca.* 50 Oe) *the direction always changed away from the direction of the present earth field at the location of the sample*

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in nature. (Experimental result XI.) *This change again indicates that a viscous magnetization caused by the earth field is removed by demagnetization.*

As a result of special studies of the tertiary basalts, I will here mention the result of an investigation of an area between normally and reversely magnetized lava-groups. This result shows that *the direction of magnetization changes gradually from reversed to normal.* (Experimental result XII.)

In fig. 4 the full points represent the magnetic south pole on the southern hemisphere, the empty circles the magnetic south pole on the northern hemisphere. Each point belongs to one lava and the successive points to successive lava flows. It is clearly seen that the change is gradual from the southern hemisphere, over the equator, and to the geographic north pole. The clockwise traces of the variation remind one of the present clockwise traces of the secular variation. Perhaps this similarity indicates that the reversal of the field took place during a period of 1000–3000 years. (Experimental result XIII.) It is difficult to understand how such changes could be caused by some self-reversal during the cooling process or during the time passed, as we would then expect random variations in the direction of magnetism. These lavas were all weakly magnetized. *The fine-structure of these changes* was only obtained after the demagnetization (before the partially demagnetization the circle of confidence is greater than after it). The intensity is about five times smaller than is otherwise the case in tertiary basalts.

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